AN ALPHA-UNIVERSAL CLASS CONTAINS EVERY STRUCTURE WHOSE SCOTT SENTENCE IS IN THE PI-ALPHA CLASS

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Editorial note. The counted statement is the author environment at source0.tex lines 795-798 and its proof environment opens at line 799, the line immediately after; the pairing is by adjacency and the retained proof range 799-804 is the whole environment.

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Editorial scope. One original result of the pinned author manuscript, counted once: an alpha-universal class contains all structures of the larger class with a P_infinity alpha Scott sentence (author label lem:universalrankinvariant).

Allowed standard prerequisites (editorial).

- Alpha-universality. K_0 is alpha-universal for K_1 when every structure in K_1 stands in the rank alpha back-and-forth relation to some structure in K_0.. It is the hypothesis.
- Scott sentences in a complexity class. A structure has a P_infinity alpha Scott sentence when its Scott sentence lies in that class, which forces it to be characterised up to isomorphism among the structures it compares with at rank alpha.. It is the property of the structure shown to lie in K_0.

Lemma 0.1. *If K_0 is α-universal for K_1, then K_0 contains all structures in K_1 with Π^{in}_α Scott sentences.*

Proof. Suppose otherwise and consider $\mathcal{A} \in K_1 \setminus K_0$ with a Π^{in}_α Scott sentence. By α-universality, there is $\mathcal{B} \in K_0$ with $\mathcal{A} \leq_\alpha \mathcal{B}$. But then $\mathcal{A} \cong \mathcal{B}$ and thus $\mathcal{A} \in K_0$, a contradiction. □

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